A Supplement for "Simple Yet Efficient Algorithms for Maximum Inner Product Search via Extreme Order Statistics"

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1 Comparison with Gaussian RP

Let $X_{[1]}$ be the concomitant of the first (maximum) order statistic $Q_{(1)}$. We recap Lemma 1.1 and Lemma 2.3 from the paper.

Lemma 1.1: Let $\theta_{\mathbf{xq}}$ be the angle between the vectors $\mathbf{x}, \mathbf{q} \in \mathbb{R}^d$. Given $0 < \epsilon < 1$ and a Gaussian random matrix $\mathbf{R} \in \mathbb{R}^{l \times d}$, we have:

$$\mathbf{Pr}\left[\left|\frac{(\mathbf{Rx})^\top \mathbf{Rq}}{l} - \mathbf{x}^\top \mathbf{q}\right| \geq \epsilon \mathbf{x}^\top \mathbf{q}\right] \leq 2\exp\left(-\frac{l}{8}\epsilon^2 \cos^2(\theta_{\mathbf{xq}})\right).$$

Lemma 2.3:

$$\mathbf{E}\left[X_{[1]}\right] \xrightarrow{D} \mathbf{x}^\top \mathbf{q}\sqrt{2\log D}\,,\ \mathbf{Var}\left[X_{[1]}\right] \xrightarrow{D} \|\mathbf{x}\|^2 - \left(\mathbf{x}^\top \mathbf{q}\right)^2.$$

Let $Z_{ceo} = X_{[1]}/\sqrt{2\log D}$, hence $\mathbf{E}\left[Z_{ceo}\right] \xrightarrow{D} \mathbf{x}^\top \mathbf{q}$ and $\mathbf{Var}\left[Z_{ceo}\right] \xrightarrow{D} \left(\|\mathbf{x}\|^2 - \left(\mathbf{x}^\top \mathbf{q}\right)^2\right)/2\log D$. We use the Chernoff bounds to bound the relative estimate error induced by $X_{[1]}$ as follows.

Chernoff bounds: Let $Z \sim N(\mu, \sigma)$, for all $t \geq 0$ we have

$$\mathbf{Pr}\left[Z \geq \mu + t\right] \leq \exp\left(-t^2/2\sigma^2\right),$$
$$\mathbf{Pr}\left[Z \leq \mu - t\right] \leq \exp\left(-t^2/2\sigma^2\right).$$

Applying the Chernoff bounds for the random variable Z_{ceo}, given any $\epsilon \geq 0$ we have:

$$\begin{aligned}
&\mathbf{Pr}\left[|Z_{ceo} - \mathbf{x}^\top \mathbf{q}| \geq \epsilon\, \mathbf{x}^\top \mathbf{q}\right] \leq 2\exp\left(\frac{-\epsilon^2 \log D\ \left(\mathbf{x}^\top \mathbf{q}\right)^2}{\|\mathbf{x}\|^2 - \left(\mathbf{x}^\top \mathbf{q}\right)^2}\right) \\
&= 2\exp\left(-\epsilon^2 \log D \cos^2(\theta_{\mathbf{xq}})/\sin^2(\theta_{\mathbf{xq}})\right) \\
&= 2D^{-\epsilon^2 \cos^2(\theta_{\mathbf{xq}})/\sin^2(\theta_{\mathbf{xq}})}.
\end{aligned}$$

When D is sufficiently large, the error probability will be arbitrarily small, and hence the estimate will be very accurate. Using the asymptotic distribution of $X_{[1]}$, our concentration bound is similar to Lemma 1.1 with $l = 8\log D/\sin^2(\theta_{\mathbf{xq}})$ projections.

2 Proof of Equation 3 and CEOs-TA pseudo-code

We consider $X_{[1]}$ and $Y_{[1]}$ and denote $\rho_{[1]}$ be the correlation between $X_{[1]}$ and $Y_{[1]}$. Let $\tau_1 = \mathbf{x}^\top \mathbf{q}$, $\tau_2 = \mathbf{y}^\top \mathbf{q}$ and assume $\tau_1 > \tau_2$. By the asymptotic property of concomitants of extreme order statistics:

$$X_{[1]} \xrightarrow{D} N\left(\tau_1\sqrt{2\log D}, \|\mathbf{x}\|^2 - \tau_1^2\right), Y_{[1]} \xrightarrow{D} N\left(\tau_2\sqrt{2\log D}, \|\mathbf{y}\|^2 - \tau_2^2\right).$$

Algorithm 1 CEOs-TA

function INDEXING(data matrix $\mathbf{X}_{d\times n}$, random Gaussian matrix $\mathbf{R}_{D\times d}$)
  Up-project $\mathbf{X}$ into D dimensions by computing $\mathbf{X}' = \mathbf{RX}$
  For each projection $i \in [D]$, sort $\mathbf{X}'_i$ and keep it together with the point indexes on the list $\mathcal{L}_i$
  return D lists $\mathcal{L}_i$ where $i \in [D]$ as our index
end function

procedure QUERYING(query point $\mathbf{q} \in \mathbb{R}^d$, data matrix $\mathbf{X}_{d\times n}$, random Gaussian matrix $\mathbf{R}_{D\times d}$, our index)
  Up-project $\mathbf{q}$ into D dimensions by computing $\mathbf{q}' = \mathbf{Rq}$
  Compute $s = 2s_0$ projections corresponding to s_0 maximum and minimum values of $\mathbf{q}'$
  Apply the TA algorithm over these s projections to compute top-b indexes with the largest inner product estimators
  Post-processing: Compute b inner products provided by the TA algorithm and return top-k points with the largest value
end procedure

Let $\sigma_{[1]}^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \tau_1^2 - \tau_2^2 + 2\rho_{[1]}\sqrt{\|\mathbf{x}\|^2 - \tau_1^2}\sqrt{\|\mathbf{y}\|^2 - \tau_2^2}$. Since the sum of two Gaussian variables is Gaussian, we have

$$Y_{[1]} - X_{[1]} \xrightarrow{D} N\left((\tau_2 - \tau_1)\sqrt{2\log D}, \sigma_{[1]}^2\right).$$

Applying Chernoff bounds for the Gaussian variable $Y_{[1]} - X_{[1]}$:

$$\begin{aligned}
&\mathbf{Pr}\left[Y_{[1]} - X_{[1]} \geq 0\right] \\
&= \mathbf{Pr}\left[Y_{[1]} - X_{[1]} \geq \mathbf{E}\left[Y_{[1]} - X_{[1]}\right] + (\tau_1 - \tau_2)\sqrt{2\log D}\right] \\
&\leq e^{-(\tau_1-\tau_2)^2 \log D/\sigma_{[1]}^2} = D^{-(\tau_1-\tau_2)^2/\sigma_{[1]}^2}.
\end{aligned}$$

Since $-1 \leq \rho_{[1]} \leq 1$, we have $\sigma_{[1]}^2 \leq \left(\sqrt{\|\mathbf{x}\|^2 - \tau_1^2} + \sqrt{\|\mathbf{y}\|^2 - \tau_2^2}\right)^2$. Hence, we can prove Equation 3:

$$\mathbf{Pr}\left[Y_{[1]} - X_{[1]} \geq 0\right] \leq D^{-(\tau_1-\tau_2)^2/\left(\sqrt{\|\mathbf{x}\|^2-\tau_1^2}+\sqrt{\|\mathbf{y}\|^2-\tau_2^2}\right)^2}.$$

3 Structured Spinners

Figure 1 shows that the Structured Spinners $\mathbf{HD}_3\mathbf{HD}_2\mathbf{HD}_1$ can simulate well the Gaussian RP with almost the same MIPS precision provided by CEOs-Est but run $O(d/\log D)$ times faster. Even though we use the Eigen library for the fast C++ matrix-vector multiplication, Structured Spinners compute the signatures 4 times faster than the Gaussian RP on Nuswide with $d = 500$ and $D = 1024$.

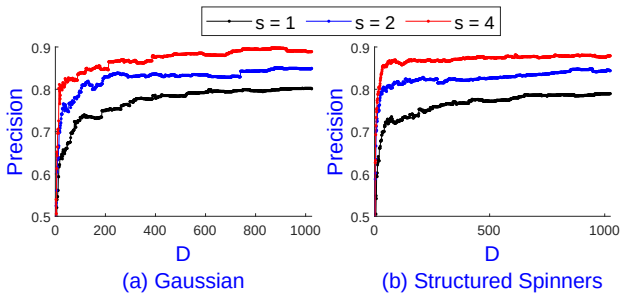

Figure 1: Top-10 MIPS precision on Nuswide with $s = \{1, 2, 4\}$ and varying D using: (a) Gaussian RP and (b) Structured Spinners to simulate Gaussian RP.

Figure 1 also reveals the accuracy of CEOs by showing the high average precision $P@10$ on top-10 MIPS when varying s and D. When fixing s and $D \geq 512$, the precision marginally increases. However, increasing s yields a substantial rise of precision. Especially, when $D = 512$ and $s = 4$, $P@10$ is nearly 90%.

4 Comparison on sublinear algorithms

Figure 2 shows that CEOs schemes are superior to LSH schemes regarding search recall on Gist and Yahoo. We observe that RangeLSH reaches the highest search recall with smaller l than SimpleLSH. Then, the accuracy of both schemes decreases when increasing l on both data sets. A simple computation indicates that the average top-1 MIPS value after the SimpleLSH transformation is 0.24 on Yahoo. Hence, the total number of collisions is at most $nL\tau_1^h \approx 1$ when $L = 512$, $h = 14$. This explains the significant small search recalls offered by the two methods.

We observe that the average top-10 MIPS values are almost the same, e.g. 0.68 on Gist, even with 1000 partitions of RangeLSH. Hence there might be no advantages of increasing the distance gap between "close" and "far way" points. Since the top-k MIPS points are often distributed on the same partition due to the similar large 2-norm values, RangeLSH reaches the highest search recall with smaller l than SimpleLSH. Since each partition will have $L/4$ hash tables, increasing l will decrease the accuracy of RangeLSH faster than SimpleLSH.

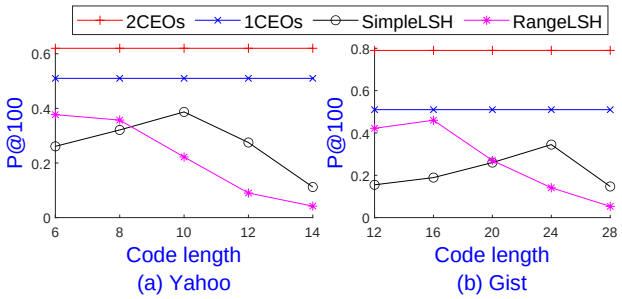

Figure 2: Comparison of top-10 MIPS accuracy of 1CEOs, 2CEOs, SimpleLSH and RangeLSH with $b = 100$ and $D = 2^{\lfloor \log d \rfloor + 1}$ on (a) Yahoo and (b) Gist.

5 Comparison on estimation algorithms

Figure 3 shows comparison between CEOs variants and LSH schemes over various k values on Gist and Msong. Although LSH schemes use $l = 128$ bit code. We observe that CEOs variants consistently achieve higher search recalls and speedup than LSH schemes.

We note that the average inner products of top-10 MIPS on Gist is 0.68 and it does not change even with 1000 partitions on RangeLSH. Their Hamming distances are approximately 14 and 28 for 64 bits and 128 bits, respectively. This means that if you use a single hash table with multi-probing, you have to probe several thousands points for inner product computations, as illustrated in the experiment in the RangeLSH paper. In contrast, CEOs variants only needs 100 inner product computations to return at least 90% recall for top-50 MIPS.

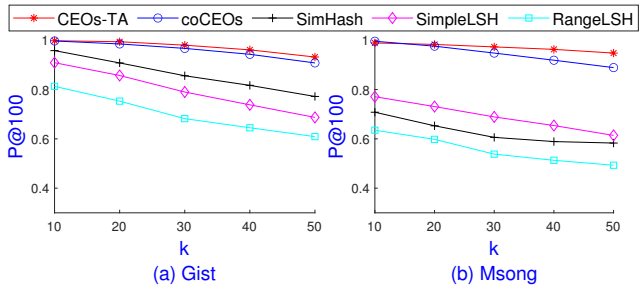

Figure 3: Comparison of top-k MIPS accuracy of CEOs variants ($B = n/100, b = 100, D = 2^{\lfloor \log d \rfloor + 1}, s = 20, s' = 40$) and LSH schemes ($l = 128$) on (a) Gist and (b) Msong.

6 Comparison with ipNSW

We show empirical results of coCEOs and ipNSW for $k = 50$. ipNSW uses $efSearch = 50$ on Yahoo and Gist, and $efSearch = 128$ on Imagenet. We observe that $efSearch = 50$ increases nearly 3 times the query time compared to $efSearch = 10$. coCEOs uses $b = 200$ for Gist and $b = 500$ for the others to achieve similar search recalls as ipNSW, as shown in Table 1. The indexing time of coCEOs is just minutes, so it would be easy to tune B and b, the key parameters to govern the query performance of coCEOs.

The computational cost of coCEOs is bounded by $2B/d + b$ inner product computations, which are $\{542, 220, 3620\}$ on Yahoo, Gist, and Imagenet, respectively. These numbers are again significantly less than that of ipNSW, i.e. $\{2674, 1697, 4065\}$ inner products for the corresponding data sets.

Table 1: Comparison of indexing time, querying time and search recall between coCEOs and ipNSW on Yahoo, Gist, and Imagenet for $k = 50$.

Dataset	coCEOs			ipNSW		
	Index	Query	Recall	Index	Query	Recall
Yahoo	**1 min**	2 ms	87%	1.8 h	2.8 ms	94%
Gist	**3 mins**	4.6 ms	91%	2.6 h	4.8 ms	95%
Imagenet	**17 mins**	9.7 ms	60%	1.7 h	2.4 ms	61%